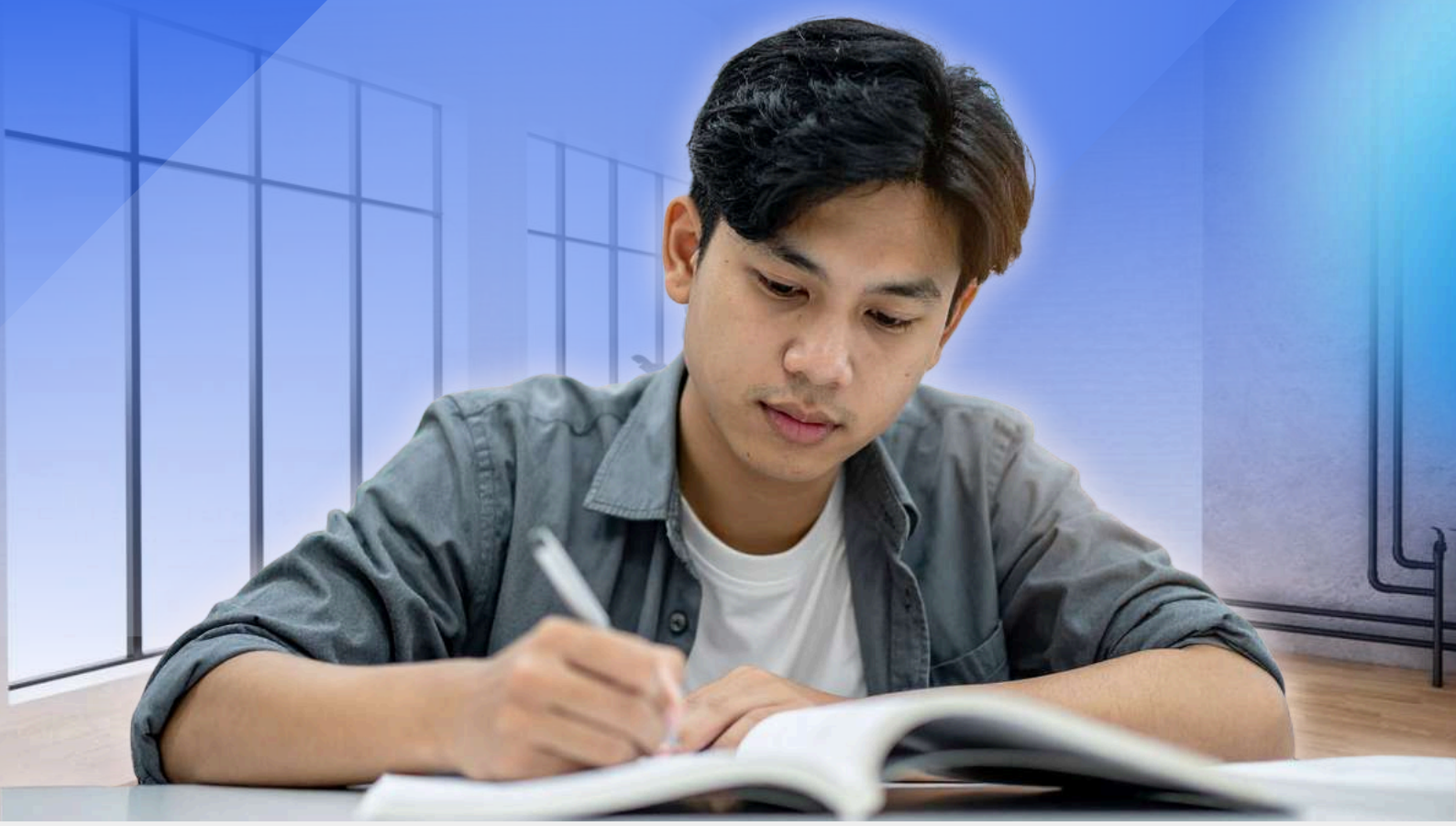
Lenterago

TRY OUT #12

SAT-Math

61012

44 QUESTIONS



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MODUL 1 & 2

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

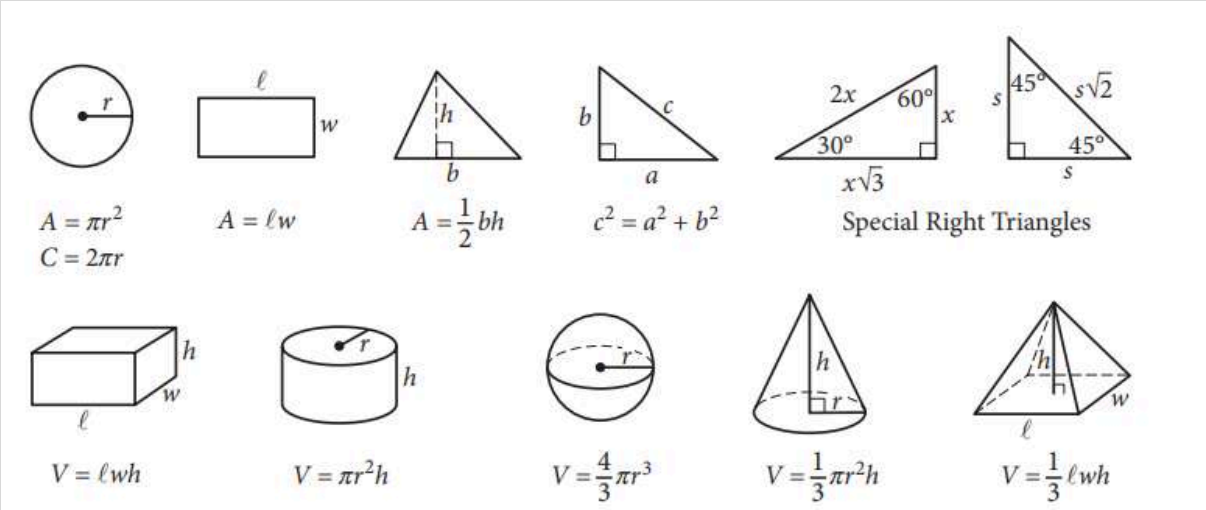
A. All variables and expressions represent real numbers.

B. Figures provided are drawn to scale.

C. All figures lie in a plane.

D. The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number

REFERENCE



The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.

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Module 1 & 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

1. Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
2. If you find more than one correct answer, write and circle only one answer.
3. Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
4. If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
5. If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
6. If your answer is a mixed number (such as 3 1/2), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
7. Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.

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NO	MATERI	SUB MATERI	KUNCI
1	Algebra	Linear Function	
	A builder uses function $f(x) = 80x + 10,000$ to estimate the cost f(x), in dollars, to build a one-story home if planned floor area of x square feet. If the builder estimates that the cost to build a certain home is $106,000, what is the planed floor area, in square feet, of the home?		
2	Algebra	Word problem of SLE	
	In the expression of $\frac{\frac{6}{5}}{\frac{12}{2y} - \frac{5}{y}} = 1$ what is the value of y?		
3	Algebra	Line	
	$y = 19.99 + 1.50x$ The equation above models the total cost y, in dollars, that a company charges a customer to rent a truck for one day and drive the truck x miles. The total cost consists of a flat fee plus a charge per mile driven. When the equation is graphed in the xy-plane, what does the y-intercept of the graph represent in terms of the model? a. A flat fee of $19.99 b. A charge per mile of $1.50 c. A charge per mile of $19.99 d. Total daily charges of $21.49		
4	Algebra	Word Problem	
	Molly bought a book and bag at a department store. The sum of the prices of the both items before sales tax was $130. There was no sales tax on the book but a 9% sales tax on the bag. The total Molly paid, including the sales tax was $136.75. What was the price, in dollars, of the book?		
5	Algebra	Word problem	
	For a certain reservoir, the function f gives the water level $f(n)$, to the nearest whole percent capacity, on the nth day of 2026. Which if the following is the best interpretation of $f(37) = 70$? a. The water level of the dam was 37% capacity for 70 days in 2026 b. The water level of the dam was 70% capacity for 37 days in 2026 c. On the 37th day of the 2026, the water level of the dam was at 70% capacity d. On the 70th day of 2026, the water level of the dam was at 73% capacity		
6	Algebra	Inequality Word Problem	
	Daniel is developing a weight loss regimen, which includes both a workout plan and a calorie-restriction plan. Daniel wants to work out for no less than 30 minutes and no more than 60 minutes a day and consume no less than 2,000 and no more than 2,500 calories. If each minute, m, of his workout time burns 50 calories, which of the following inequalities represents the number of minutes, m, that Daniel can work out each dayto burn off at least as many calories as he consumes? a. $30 \leq m \leq 60$ b. $30 \leq m \leq 50$ c. $40 \leq m < 50$ d. $40 \leq m \leq 50$		
7	Algebra	Linear Function	

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A group of 202 people went on an overnight camping trip, taking 60 tents with them. Some of the tents held 2 people each, and the rest held 4 people each. Assuming all the tents were filled to capacity and every person got to sleep in a tent, exactly how many of the tents were 2-person tents?

a. 30
b. 20
c. 19
d. 18

8	Advanced Math	Quadratic Word Problem	

A car is traveling at x speed per second. The driver sees a red light ahead and after 1.5 second reaction time, the driver applies the brake. After the brake is applied, the car takes $\frac{x}{24}$ second to stop, during which time the average speed of the car is $\frac{x}{2}$ feet per second. If the car travel 165 feet from the time the driver saw the red light to the time it comes to a complete stop, which of the following equation can be used to find the value of x?

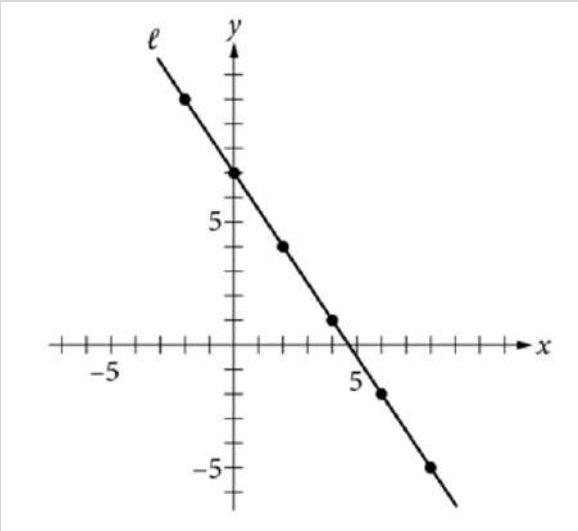
a. $x^2 + 48x - 3,960 = 0$
b. $x^2 + 48x - 7,920 = 0$
c. $x^2 + 72x - 3,960 = 0$
d. $x^2 + 72x - 7,920 = 0$

9	Advanced Math	Quadratic Equation	

The graph of function g in the xy plane is a parabola defined by $g(x) = (x - 2)(x - 5)$. Which of the following interval contains the x −coordinate of vertex of the graph?

a. $6 < x < 8$
b. $4 < x < 6$
c. $-2 < x < 4$
d. $-4 < x < -2$

10	Advanced Math	Transformation of function	



If line l is translated up 5 units and right 7 units, what is the slope if the new line?

a. $\frac{2}{5}$

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b. $-\frac{3}{2}$

c. $-\frac{8}{9}$

d. $-\frac{11}{4}$

11	Advanced Math	Rational Expression	

Which of the following is equivalent to $\frac{2m+6}{4} \times \frac{6m-36}{3m+9}$?

a. $\frac{12m^2-216}{12m+36}$

b. $\frac{8m-30}{3m+13}$

c. $\frac{m-6}{4}$

d. $m - 6$

12	Advanced Math	Exponent and Radical	

A gas in a container will escape through holes of microscopic size as long as the holes are larger than the gas molecules. The process is called effusion. If a gas of molar mass M_1 effuses at a rate of r_1 and a gas of molar M_2 effuses at rate of r_2, then the following relationship holds (called Graham's Law)

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

Which of the following correctly express M_2 in the term of M_1, r_1 and r_2?

a. $M_2 = M_1\left(\frac{r_1^2}{r_2^2}\right)$

b. $M_2 = M_1\left(\frac{r_2^2}{r_1^2}\right)$

c. $M_2 = \sqrt{M_1}\left(\frac{r_1}{r_2}\right)$

d. $M_2 = \sqrt{M_1}\left(\frac{r_2}{r_1}\right)$

13	Advanced Math	Exponential Word Problem	

Time (hours)	Number of Bacteria
0	1.0×10^3
1	4.0×10^3
2	1.6×10^4
3	6.4×10^4

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The table above gives the initial number (at time $t = 0$) of bacteria placed in a growth medium and the number of bacteria in the growth medium over 3 hours. Which of the following function best models the number of bacteria , $N(t)$, after t hours?

a. $N(t) = 4,000t$
b. $N(t) = 1,000 + 3,000t$
c. $N(t) = 1,000(4^{-t})$
d. $N(t) = 1,000(4^{t})$

14	Advanced Math	Polynomial	

$$y^5 - 2y^4 - cxy + 6x$$

In the polynomial above, c is a constant. If the polynomial is divisible by $y - 2$ what is the value of c?

15	Problem Solving and Data Analysis	Rate and Percent	

Dr. Valen, a noted dietician, mixes different solutions as part of her research into sugar substitutes. By weight, she mixes 40% of sample A with 70% of a sample B to create substitute C. If Dr. Valen initially had 60 grams of substitute A and 110 grams of substitute B, then what would be the weight, in grams, of substitute C ?

a. 24
b. 77
c. 101
d. 170

16	Problem Solving and Data Analysis	Rate	

The scientists estimate that the Pacific Plate has move about 1,060 kilometers in the past 10.3 million years. What was the average speed of the Pacific Plate during that time period, in centimeters per year?

a. 1.03
b. 10.3
c. 103
d. 1,030

17	Problem Solving and Data Analysis	Statistic	

Ages of 20 student Enrolled in a College Class

Age	Frequency
18	6
19	5
20	4
21	2
22	1
23	1
30	1

The table above shows the distribution of ages of the 20 students enrolled in a college class. Which of the following gives the mean, median and mode of the ages?

a. Mode < median < mean
b. Mode < mean < median
c. Median < mode < mean

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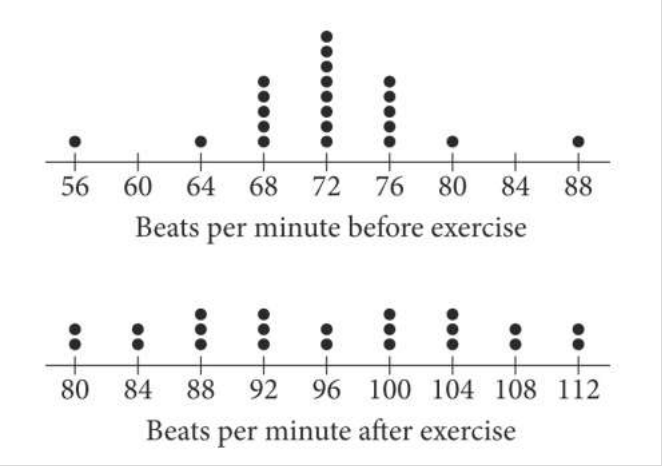
d. Mean < mode < median

18	Problem Solving and Data Analysis	Standard Deviation of Data	

The 22 students in a health class conducted an experiment in which they each recorded their pulse rates, in beats per minute, before and after completing a light exercise routine. The dot plots below display the results.



Let s_1 and r_1 be the standard deviation and range, respectively, of the data before exercise, and let s_2 and r_2 be the standard deviation and range, respectively, of the data after exercise. Which of the following is true?

a. $s_1 = s_2$ and $r_1 = r_2$
b. $s_1 < s_2$ and $r_1 < r_2$
c. $s_1 > s_2$ and $r_1 > r_2$
d. $s_1 < s_2$ and $r_1 = r_2$

19	Problem Solving and Data Analysis	Probability	

Dream Recalled during One Week

	None	1 to 4	5 or more	Total
Group X	15	28	57	100
Group Y	21	11	68	100
Total	36	39	125	200

The data in the table above were produced by a sleep researcher studying the number of dream people recalled when asked to record their dream for one week. Group X consisted of 100 people who observed early bedtimes and Group Y consisted of 100 people who observed later bedtimes. If a person is chosen at random from those who recalled at least 1 dream, what is the probability that person belonged to Group Y?

a. $\frac{68}{100}$
b. $\frac{79}{100}$
c. $\frac{79}{164}$
d. $\frac{164}{200}$

20	Geometry	Area of Circle	

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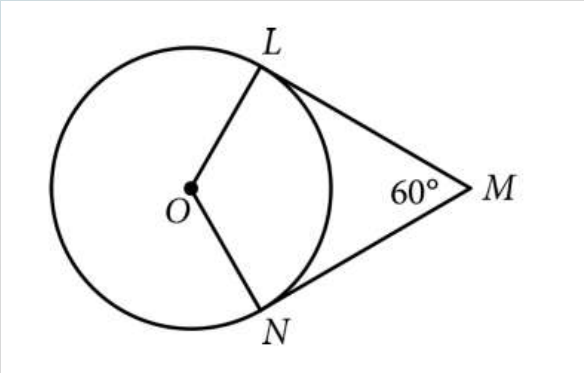


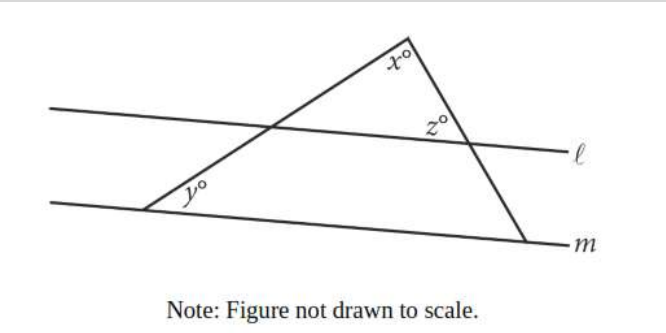
In the figure above, point O is the center of the circle, line segments LM and MN are tangent to the circle at points L and N, respectively. The segment intersects at point M as shown. If the circumference of the circle is 96, what is the length of the minor arc LN?

21	Geometry	Volume	

A cylindrical can containing pieces of fruit is filled to the top with syrup before being sealed. The base of the can has an area of 75 cm^2 and the height of the can is 10 cm. If the 100 cm^3 of syrup is needed to fill the can to the top, which of the following is closest to the total volume of the fruit pieces in the can?

a. 7.5 cm^3
b. 185 cm^3
c. 640 cm^3
d. 750 cm^3

22	Geometry	Line and Angle	



In the figure above line l and m are parallel, $y = 20$ and $z = 60$. What is the value of x?

a. 120
b. 100
c. 90
d. 80

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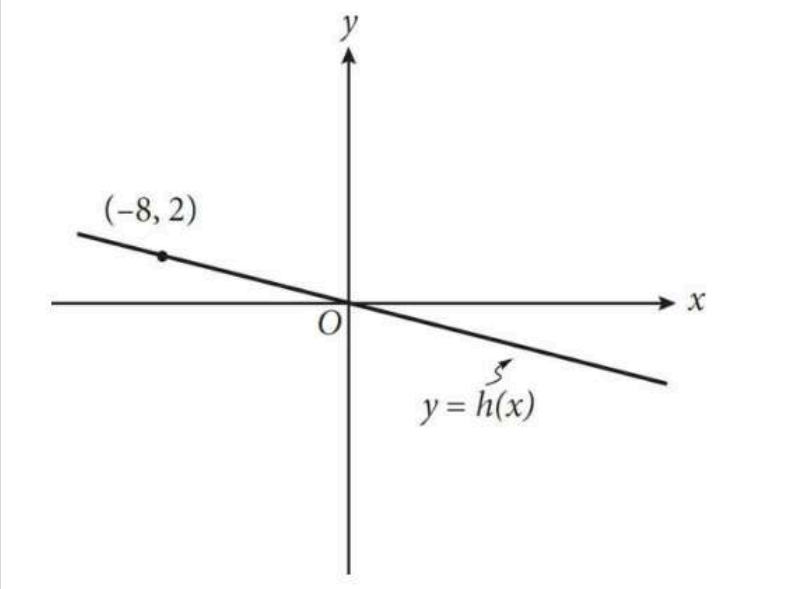
NO	MATERI	SUB MATERI	KUNCI
1	Algebra	Inequality	
	If $3p - 2 > 1$ and p is whole number . What is the least possible value of 6p+5? a. 11 b. 13 c. 15 d. 17		
2	Algebra	Slope of linear equation	
	 In the xy-plane above, appoint (not shown) with coordinate (a, b) lies on the graph of the linear equation h. If a and b are nonzero integer, what is the ratio of b to a? a. -4 to 1 b. -2 to 1 c. -1 to 2 d. -1 to 4		
3	Algebra	Word Problem	

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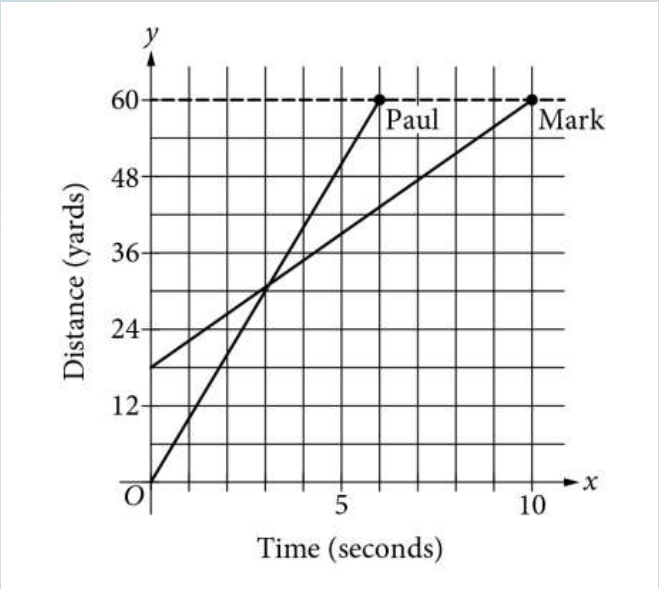

The graph above shows the positions of Paul and Mark during a race. Paul and Mark each ran at a constant rate, and Mark was given a head start to reduce the distance he needed to run. Paul finished the race in 6 seconds, and Mark finished the race in 10 seconds. According to the graph, Mark was given a head start of how many yards?

a. 3
b. 12
c. 18
d. 24

4	Algebra	Linear Equation word problem	
	How many liters of a 25% saline solution must be added to 3 liters of a 10% saline solution to obtain a 15% saline solution?		
5	Algebra	Inequality	

$$-2y \leq 8$$
$$y - 3 \leq x$$
$$-\frac{1}{3}y + 1 \geq x$$

Which of the following graph show the solution to the system of inequalities above?

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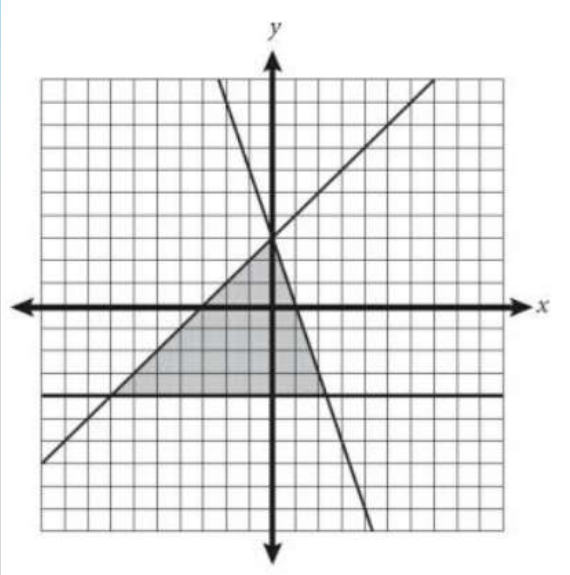
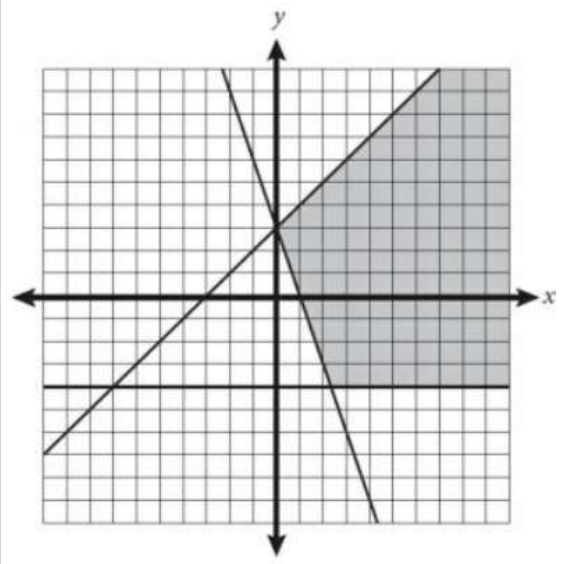
a.



b.



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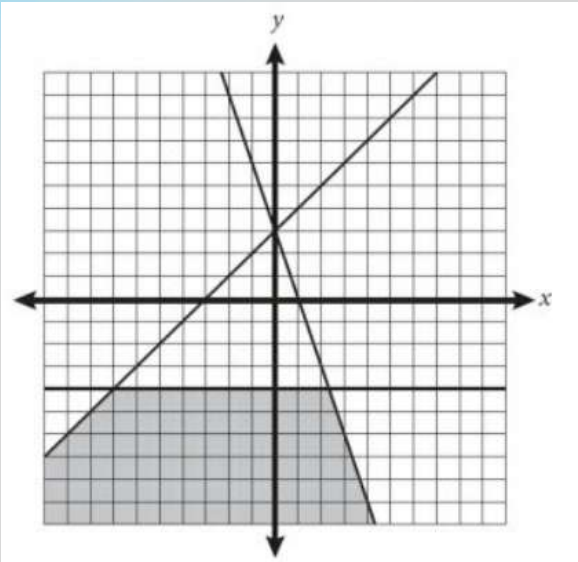
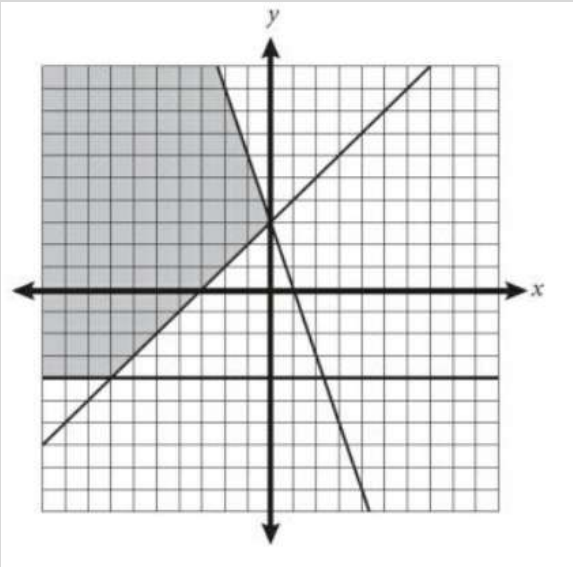
c.



d.



6	Algebra	Inequality Word problem	

Each morning Peter jogs at 5 miles per hour and rides a bike at 12 miles per hour. His goal is to jog and ride his bike a total of at least 9 miles in no more than 2 hours. If Peter jogs j miles and rides his bike b miles, which of the following system of inequalities represent Peter's goals?

a. $\frac{j}{5} + \frac{b}{12} \leq 2$
$j + b \geq 9$

b. $5j + 12b \leq 9$
$j + b \leq 2$

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c. $\frac{j}{5} + \frac{b}{12} \geq 2$
$j + b \leq 9$

d. $5j + 12b \geq 9$
$j + b \geq 2$

7	Algebra	System of Linear Equation	

$$x + 7y =- 10$$
$$3x - 4y = k$$

In the system of equations above, k is a constant. If (a, b) is the solution to the system, what is the value of a, in the term of k?

a. $\frac{-k-30}{25}$
b. $\frac{3k+10}{25}$
c. $\frac{6k-8}{25}$
d. $\frac{7k-40}{25}$

8	Advanced	Function	

$$g(t) = \frac{5(7t-12c)}{2} - 25$$

The number of people who go to a public swimming pool can be modeled by the function g above, where c is a constant and t is the air temperature in degrees Fahrenheit (°F) for $70 < t < 100$. If 350 people are predicted to go to the pool when the temperature is 90°F, what is the value of c?

a. 20
b. 40
c. 60
d. 80

9	Advanced Math	Polynomial and radical	

$$\frac{2d^2-d-10}{d^2+7d+10} = \frac{d^2-4d+3}{d^2+2d-15}$$

In the equation above, what is the value of d?

a. -4
b. 2
c. 4
d. 6

10	Advanced Math	System of Quadratic Equation	

$$y = x^2 + 3x - 7$$
$$y - 5x + 8 = 0$$

How many solutions are there to the system of equation above?

a. There are exactly 4 solutions
b. There are exactly 2 solutions

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c. There is exactly 1 solution
d. There is no solution

11	Advanced Math	Rational Expression	

$$f(x) = x^3 - 9x$$
$$g(x) = x^2 - 2x - 3$$

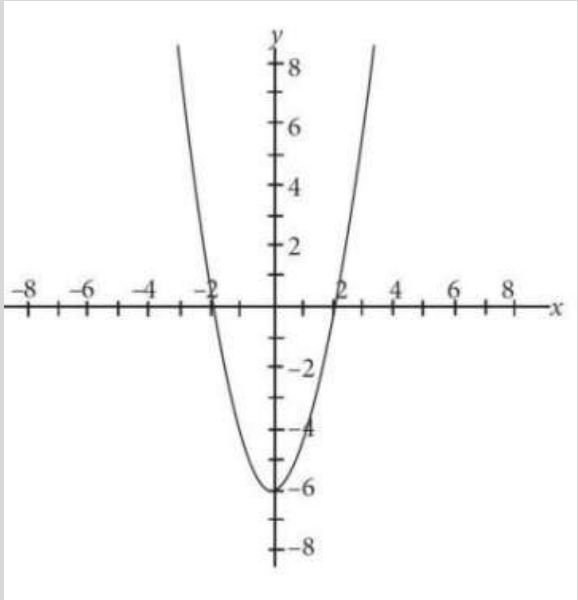
Which of the following expression is equivalent to $\frac{f(x)}{g(x)}$, for $x > 3$?

a. $\frac{1}{x+1}$
b. $\frac{x+3}{x+1}$
c. $\frac{x(x-3)}{x+1}$
d. $\frac{x(x+3)}{x+1}$

12	Advanced Math	Polynomial	

A certain polynomial, P, has a degree of 2. Polynomial P has zeros of 2 and –3, and a > 0 when the function of polynomial P is written in the form of $y = ax^2 + bx + c$ Given this information, which of the following could be the graph of polynomial P ?

a.



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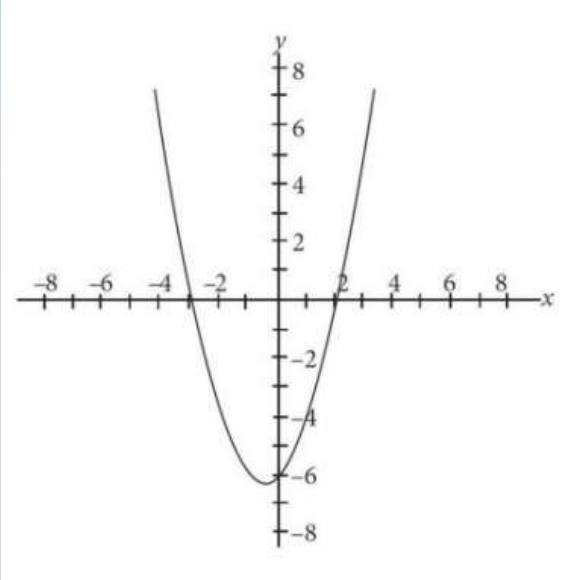
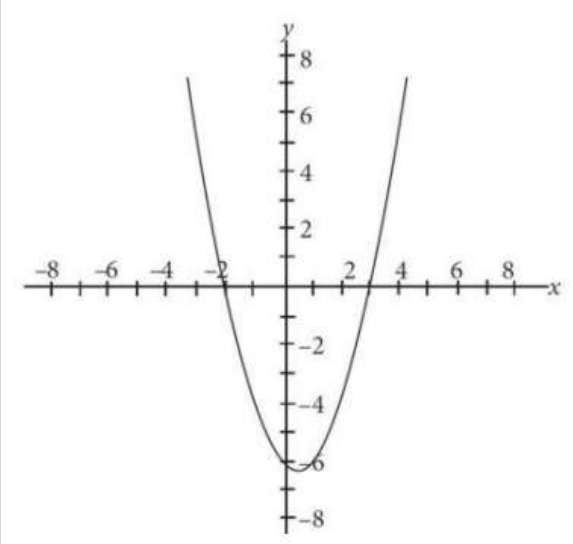


c.



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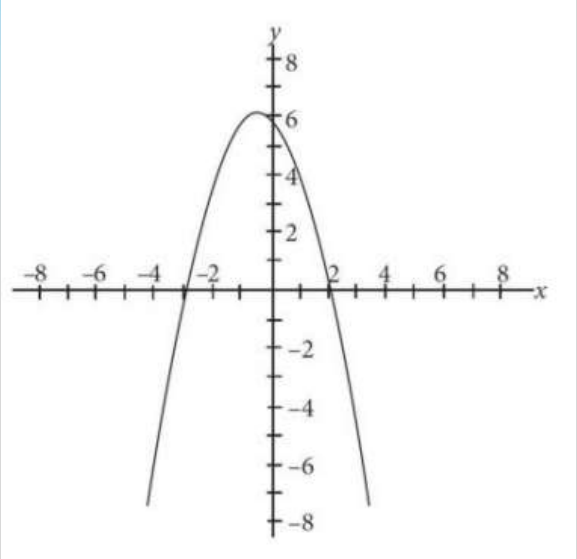
d.



13	e. Advanced Math	Function transformation	

$$x^2 + (y + 1)^2 = 4$$

The graph of the equation above in the xy-plane is a circle. If the center of the circle translated 1 unit up and radius is increased by 2, which the following is an equation of the resulting circle?

a. $x^2 + y^2 = 6$
b. $x^2 + y^2 = 16$
c. $x^2 + (y + 2)^2 = 6$
d. $x^2 + (y + 2)^2 = 16$

14	Advanced Math	Polynomial	

Which of the following is equivalent to $16s^4 - 4t^2$?

a. $4(s^2 - t)(4s^2 + t)$
b. $4(s4^2 - t)(s^2 + t)$
c. $4(2s^2 - t)(2s^2 + t)$
d. $(8s^2 - 2t)(2s^2 + 2t)$

15	Problem Solving and Data Analysis	Rate and Unit	

A furniture store buys its furniture from a wholesaler. For a table, the store usually charges its cost from the wholesaler plus 75%. During a sale, the store charged the cost from the wholesaler plus 15%. If the sale price of the table is $299, what is the usual price for the table?

a. $359
b. $455
c. $479
d. $524

16	Problem Solving and Data Analysis	Probability	

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		Flavor	
		Vanilla	Chocolate
Topping	Hot fudge	8	6
	Caramel	5	6

The table above shows the flavors of ice cream and the toppings chosen by the people at a party. Each person chose one flavor of ice cream and one topping. Of the people who chose vanilla ice cream, what fraction chose hot fudge as a topping?

a. $\frac{8}{25}$

b. $\frac{5}{13}$

c. $\frac{13}{25}$

d. $\frac{8}{13}$

17	Problem Solving and Data Analysis	Margin of Error	

A research assistant randomly selected 75 undergraduate students from the list of all students enrolled in the psychology-degree program at a large university. She asked each of the 75 students, "How many minutes per day do you typically spend reading?" The mean reading time in the sample was 89 minutes, and the margin of error for this estimate was 4.28 minutes. Another research assistant intends to replicate the survey and will attempt to get a smaller margin of error. Which of the following samples will most likely result in a smaller margin of error for the estimated mean reading time per day for students in the psychology-degree program?

a. 40 randomly selected undergraduate psychology-degree program students
b. 40 randomly selected undergraduate students from all degree programs at the university
c. 300 randomly selected undergraduate psychology-degree program students
d. 300 randomly selected undergraduate students from all degree programs at the university

18	Problem Solving and Data Analysis	Statistic	

The mean score of 8 players in a basketball game was 14.5 points. If the highest individual score is removed, the mean score of the remaining 7 players becomes 12 points. What was the highest score?

a. 20
b. 24
c. 32
d. 36

19	Geometry	Line and Angle	

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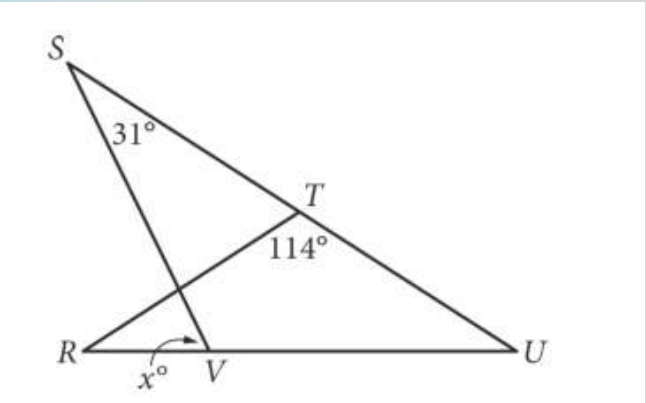
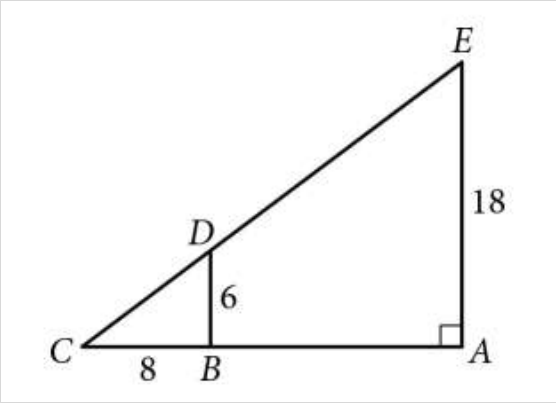
<tr><td></td><td>


In the figure above, RT = TU. What is the value of x?
a. 72
b. 66
c. 64
d. 58
</td></tr>
<tr><td rowspan="2">20</td><td>Geometry</td><td>Triangle</td><td></td></tr>
<tr><td colspan="3">


In the figure above, BD is parallel to AE. What is the length of CE?
</td></tr>
<tr><td rowspan="2">21</td><td>Geometry</td><td>Circle</td><td></td></tr>
<tr><td colspan="3">
$$(x - 6)^2 + (y + 5)^2 = 16$$
In the xy-plane, the graph of equation above is a circle. Point P is on the circle and has coordinates $(10, -5)$. If PQ is a diameter of the circle, what are the coordinates of point Q?
a. $(2, -5)$
b. $(6, -1)$
c. $(6, -5)$
d. $(6, -9)$
</td></tr>
<tr><td>22</td><td>Geometry</td><td>Area</td><td></td></tr>
</table>

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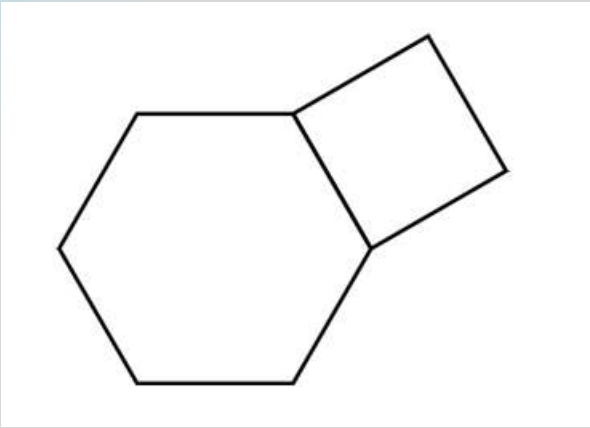

The figure above shown a regular hexagon with sides length a and a square with length sides a. If the area of hexagon is $384\sqrt{3}$ square inches, what is the area, in square inches, of the square?

A. 256
B. 192
C. $64\sqrt{3}$
D. $16\sqrt{3}$

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